Statistical Randomness Assessment of 7 Variables (Y, T, W, R, ABC, Q, E)

Structured report · N = 328 rows × 7 variables · Tests and thresholds defined in advance by the requester · Every number below is unmodified program output (results/*.csv)

Overall result against the pre-defined criterion "pass 100% of the randomness tests": NOT MET.
NIST SP 800-22 battery: PASS 0/15 · FAIL 9/15 · N/A (sequence too short for the standard) 6/15. Chi-square uniformity: 1/7 variables pass. Shannon entropy ≥ 99.9%: 0/7 variables.

Layer 3: Verbal scientific summary (Hebrew)

שאלת המחקר: האם 7 המשתנים בטבלה המאוחדת (Y, T, W, R, ABC, Q, E; N=328 שורות) עוברים את כל מבחני האקראיות שהוגדרו מראש (אחידות, אי-תלות, אנטרופיה וסוללת NIST SP 800-22), כפי שנדרש כדי לקבוע שאלה "נתוני הגרלה"?

תשובה: לא. הנתונים **אינם עומדים** בקריטריון שנקבע מראש (מעבר של 100% מהמבחנים).

- **אחידות:** מבחן χ^2 דוחה אחידות ב-6 מתוך 7 המשתנים (p בין 10^{-97} ל-10^{-14}). רק ABC עובר (χ^2 p=0.2319, KS p=0.7192). יחס המעבר הוא 0.1429, הרחק מתחת ל-0.99. ההיסטוגרמות אינן שטוחות, למעט ABC.
- **אי-תלות (על הערכים המקוריים):** רוב מבחני הרצפים עוברים את הסף p>0.05. יש שני כישלונות: Q במבחן רצפי עלייה/ירידה (p=0.0086), ו-R במבחן רצפים מעל/מתחת לחציון (p=0.0490). כל מקדמי האוטו-קורלציה קטנים (|r|≤0.161). 10 מתוך 140 מקדמים חורגים מרצועת 95%, וכל העמודות נמצאות בתוך הרצועה רק עבור W, ABC.
- **אנטרופיה:** אף משתנה לא מגיע ל-99.9% מהאנטרופיה המקסימלית. הקרוב ביותר הוא ABC עם 0.9938.
- **NIST SP 800-22:** PASS 0/15, FAIL 9/15, N/A 6/15. שישה מבחנים לא ניתנים להרצה, כי אורך הרצף (10,824 ביטים ברצף המאוחד) קטן מהמינימום שהתקן דורש.

הערות אפיסטמולוגיות (אינן משנות את ההכרעה ואינן מחליפות ספים):

1. "עבר מבחן" אינו הוכחה לאקראיות. הוא רק אי-דחייה של השערת האפס במבחן מסוים. לכן גם 15/15 לא היה "מוכיח" אקראיות. לעומת זאת, דחייה חזקה ($p\approx10^{-97}$) היא ראיה חזקה נגד אחידות.
2. תחום האחידות הוגדר כטווח השלמים מהמינימום ועד המקסימום שנצפו בכל משתנה. התחום התיאורטי האמיתי לא היה ידוע.
3. המרת הערכים לביטים (value−min ב-$\lceil \log_2 k \rceil$ ביטים) מטה את תדירות האחדות כאשר k אינו חזקה של 2. זה קורה גם לנתונים אחידים לחלוטין (ראו עמודת ref בטבלת האנטרופיה). לכן כישלונות NIST על הביטים משקפים גם את הקידוד, ולא רק את הנתונים.
4. עבור N=328, אומדן האנטרופיה הנאיבי מוטה כלפי מטה. גם לנתונים אחידים לחלוטין היחס הצפוי הוא כ-0.987–0.995 (Miller–Madow), כלומר סף 99.9% כמעט אינו בר-השגה במדגם בגודל הזה. הסף לא שונה, והתוצאה מוצגת כפי שהיא.
5. מבחן ה-DFT: המימוש לא שחזר את דוגמת המספרים שבמסמך התקן (ראו נספח אימות). כל שאר דוגמאות התקן שנבדקו (12/13) שוחזרו בדיוק של 6 ספרות.
6. תיקון שבוצע במהלך העבודה: הריצה הראשונה של מבחן KS השתמשה ב-scipy.kstest, שנכתב לנתונים רציפים, ולכן ניפח את D בנתונים בדידים עם ערכים חוזרים. המבחן הוחלף ב-KS הבדיד הנכון (Conover 1972). הדבר שינה את תוצאת ABC בלבד (מ-$p\approx7.6\cdot10^{-10}$ ל-p=0.719). χ^2 לא הושפע.
7. ערכי ה-p ויחסי המעבר שהופיעו בבקשה כ"צפויים" (למשל 0.4129 או 0.9885) לא שימשו בשום שלב. בכל מקום מוצגים רק ערכים שחושבו בפועל.

Layer 2: Tables

Checklist, item by item as specified

checked	section	item	required threshold	observed (machine output)	note	status
✔	1. Uniformity	Frequency test, chi-square, per variable	p > 0.01 (aim > 0.05)	Y=1.707e-97; T=1.124e-31; W=6.376e-15; R=1.674e-20; ABC=0.2319; Q=1.787e-74; E=1.645e-14	1/7 variables meet it	**FAIL**
✔	1. Uniformity	Frequency test, Kolmogorov-Smirnov (discrete), per variable	p > 0.01 (aim > 0.05)	Y=5.081e-76; T=1.45e-38; W=4.521e-18; R=3.655e-11; ABC=0.7192; Q=1.066e-51; E=2.954e-08	1/7 variables meet it	**FAIL**
✔	1. Uniformity	Proportion of variables passing chi-square (NIST style)	about 0.99 (NIST interval for m=7: >= 0.8772)	0.1429		**FAIL**
✔	1. Uniformity	NIST Frequency (monobit)	p > 0.01, proportion ~0.99	combined-stream p = 7.239e-17; proportion 2/7 = 0.2857; status FAIL		**FAIL**
✔	1. Uniformity	NIST Block Frequency	p > 0.01 (aim > 0.05), proportion ~0.99	combined-stream p = 0.000842; proportion 6/7 = 0.8571; status FAIL		**FAIL**
✔	1. Uniformity	Frequency histogram (visual)	relatively flat bars, slight fluctuation only	Figure 1	Only ABC looks flat. Y and T pile up at high values, Q at low values, and W, R, E are hump-shaped	**FAIL**
✔	2. Independence	Runs test, up/down runs on the original values	p > 0.05	Y=0.9625; T=0.8219; W=0.0870; R=0.3497; ABC=0.7061; Q=0.0086; E=0.4755	6/7 meet it (fails: Q)	**FAIL**
✔	2. Independence	Runs test, runs of identical symbols (above/below median)	p > 0.05	Y=0.9835; T=0.6915; W=0.7749; R=0.0490; ABC=0.3644; Q=0.3603; E=0.8211	6/7 meet it (fails: R)	**FAIL**
✔	2. Independence	NIST Runs (bits)	p > 0.05, proportion ~0.99	combined-stream p = 0.000e+00; proportion 0/7 = 0.0000; status FAIL		**FAIL**
✔	2. Independence	NIST Longest Run of Ones	p > 0.05, proportion ~0.99	combined-stream p = 2.749e-24; proportion 1/7 = 0.1429; status FAIL		**FAIL**
✔	2. Independence	Autocorrelation coefficient, lags 1-20	tends to 0 within statistical error (95% band ±1.96/√N)	Y=0.161; T=0.110; W=0.095; R=0.137; ABC=0.103; Q=0.115; E=0.141 (max |r|)	10/140 coefficients outside the band; by chance about 7/140 are expected outside a 95% band	**FAIL**
✔	2. Independence	Correlogram (visual)	all bars except lag 0 inside the confidence band	Figure 2	All bars inside the band only for: W, ABC. The other columns each have 1-3 bars outside	**FAIL**
✔	2. Independence	NIST Random Excursions	p > 0.05, proportion ~0.99	N/A — n=10824 (requires >= 1,000,000); J=10 cycles (requires >= 500)		**N/A**
✔	2. Independence	NIST Random Excursions Variant	p > 0.05, proportion ~0.99	N/A — n=10824 (requires >= 1,000,000); J=10 cycles (requires >= 500)		**N/A**
✔	3. Entropy	Shannon entropy of the values / maximum log2(k)	>= 99.9% of maximum	Y=0.7716; T=0.8849; W=0.9342; R=0.9205; ABC=0.9938; Q=0.8346; E=0.9301	meets: none	**FAIL**
✔	3. Entropy	Shannon entropy of the bit streams (max 1.000)	>= 0.999	Y=0.9919; T=0.9892; W=0.9695; R=0.9999; ABC=0.9878; Q=0.8451; E=0.9994; COMBINED=0.9954	meets: R, E	**FAIL**
✔	3. Entropy	NIST Maurer's Universal	p > 0.05, proportion ~0.99	N/A — n=10824 < 387,840		**N/A**
✔	3. Entropy	NIST Approximate Entropy	p > 0.05, proportion ~0.99	combined-stream p = 1.083e-30; proportion 0/7 = 0.0000; status FAIL		**FAIL**
✔	4. NIST SP 800-22	Binary Matrix Rank	p > 0.01, proportion ~0.99	N/A — n=10824 < 38*M*Q=38912		**N/A**
✔	4. NIST SP 800-22	Full battery, 15 tests	15/15 PASS (100%)	PASS 0/15, FAIL 9/15, N/A 6/15		**FAIL**

NIST SP 800-22 scorecard (15 tests)

Combined stream = the 7 encoded variables interleaved row by row (10,824 bits): a single p-value per test. Proportion = share of the 7 per-variable streams (984–1,640 bits) with p ≥ 0.01. The NIST acceptable proportion interval for m=7 is ≥ 0.8772, i.e. 7/7. PASS requires combined p ≥ 0.01 and a proportion inside the interval. N/A = the input-size requirement in SP 800-22 §2.x.7 is not met, so no p-value is reported.

#	test	combined_p_values	proportion	status	reason
1	Frequency	7.239e-17	2/7 = 0.2857	**FAIL**	—
2	BlockFrequency	0.000842	6/7 = 0.8571	**FAIL**	—
3	Runs	0.000e+00	0/7 = 0.0000	**FAIL**	—
4	LongestRun	2.749e-24	1/7 = 0.1429	**FAIL**	—
5	BinaryMatrixRank	—	—	**N/A**	n=10824 < 38*M*Q=38912
6	DFT	1.215e-05	3/6 = 0.5000	**FAIL**	implementation not validated against spec example (see validation); 1 column stream(s) below minimum length
7	NonOverlappingTemplate	3.719e-06	6/7 = 0.8571	**FAIL**	—
8	OverlappingTemplate	—	—	**N/A**	n=10824 < 1,000,000
9	MaurerUniversal	—	—	**N/A**	n=10824 < 387,840
10	LinearComplexity	—	—	**N/A**	n=10824 < 1,000,000
11	Serial	1.690e-21 / 0.034884	1/7 = 0.1429	**FAIL**	—
12	ApproximateEntropy	1.083e-30	0/7 = 0.0000	**FAIL**	—
13	CumulativeSums	3.337e-17 / 2.613e-17	2/7 = 0.2857	**FAIL**	—
14	RandomExcursions	—	—	**N/A**	n=10824 (requires >= 1,000,000); J=10 cycles (requires >= 500)
15	RandomExcursionsVariant	—	—	**N/A**	n=10824 (requires >= 1,000,000); J=10 cycles (requires >= 500)

1. Uniformity: chi-square and discrete KS (domain = observed min..max)

column	min	max	k	chi2	df	p_chi2	KS_D	p_KS
Y	11	37	27	540.1159	26	1.707e-97	0.4990	5.081e-76
T	7	36	30	222.2439	29	1.124e-31	0.3600	1.45e-38
W	2	28	27	125.0732	26	6.376e-15	0.2467	4.521e-18
R	6	35	30	162.4268	29	1.674e-20	0.1929	3.655e-11
ABC	1	7	7	8.0854	6	0.2319	0.0379	0.7192
Q	1	23	23	416.2744	22	1.787e-74	0.4152	1.066e-51
E	3	33	31	130.5732	30	1.645e-14	0.1648	2.954e-08

2. Independence: autocorrelation and runs tests (original values)

column	acf lag1	max abs acf lags1 20	lags outside band	runs updown R	runs updown expected	runs updown p	runs median R	runs median expected	runs median p
Y	0.0768	0.1610	2	190	189.6667	0.9625	144	143.8252	0.9835
T	0.0556	0.1101	2	208	206.3333	0.8219	150	153.4590	0.6915
W	-0.0894	0.0949	0	219	206.3333	0.0870	157	154.4984	0.7749
R	0.0615	0.1375	2	218	211.0000	0.3497	140	157.3706	0.0490
ABC	0.0859	0.1034	0	191	188.3333	0.7061	134	141.5654	0.3644
Q	-0.0726	0.1148	1	209	190.3333	0.0086	158	150.1237	0.3603
E	0.0311	0.1411	3	205	210.3333	0.4755	155	156.9936	0.8211

3. Shannon entropy

The 'ref' columns are reference values only, not tests. "ref expected H ratio" = the expected plug-in ratio if the data were exactly uniform with N=328 (Miller–Madow). "ref ones fraction" = the bit-1 frequency the binary encoding would produce for exactly uniform values.

column	k	H bits	H max	H ratio	n bits	ones fraction	H binary	ref expected H ratio if uniform N328	ref ones fraction if exactly uniform
Y	27.0000	3.6690	4.7549	0.7716	1640	0.5530	0.9919	0.9880	0.4444
T	30.0000	4.3421	4.9069	0.8849	1640	0.5610	0.9892	0.9870	0.4733
W	27.0000	4.4421	4.7549	0.9342	1640	0.3976	0.9695	0.9880	0.4444
R	30.0000	4.5167	4.9069	0.9205	1640	0.5049	0.9999	0.9870	0.4733
ABC	7.0000	2.7899	2.8074	0.9938	984	0.4350	0.9878	0.9953	0.4286
Q	23.0000	3.7756	4.5236	0.8346	1640	0.2726	0.8451	0.9893	0.4174
E	31.0000	4.6081	4.9542	0.9301	1640	0.4854	0.9994	0.9867	0.4839
COMBINED	—	—	—	—	10824	0.4599	0.9954	—	—

Layer 1: Graphics

Figure 1: Frequency histograms

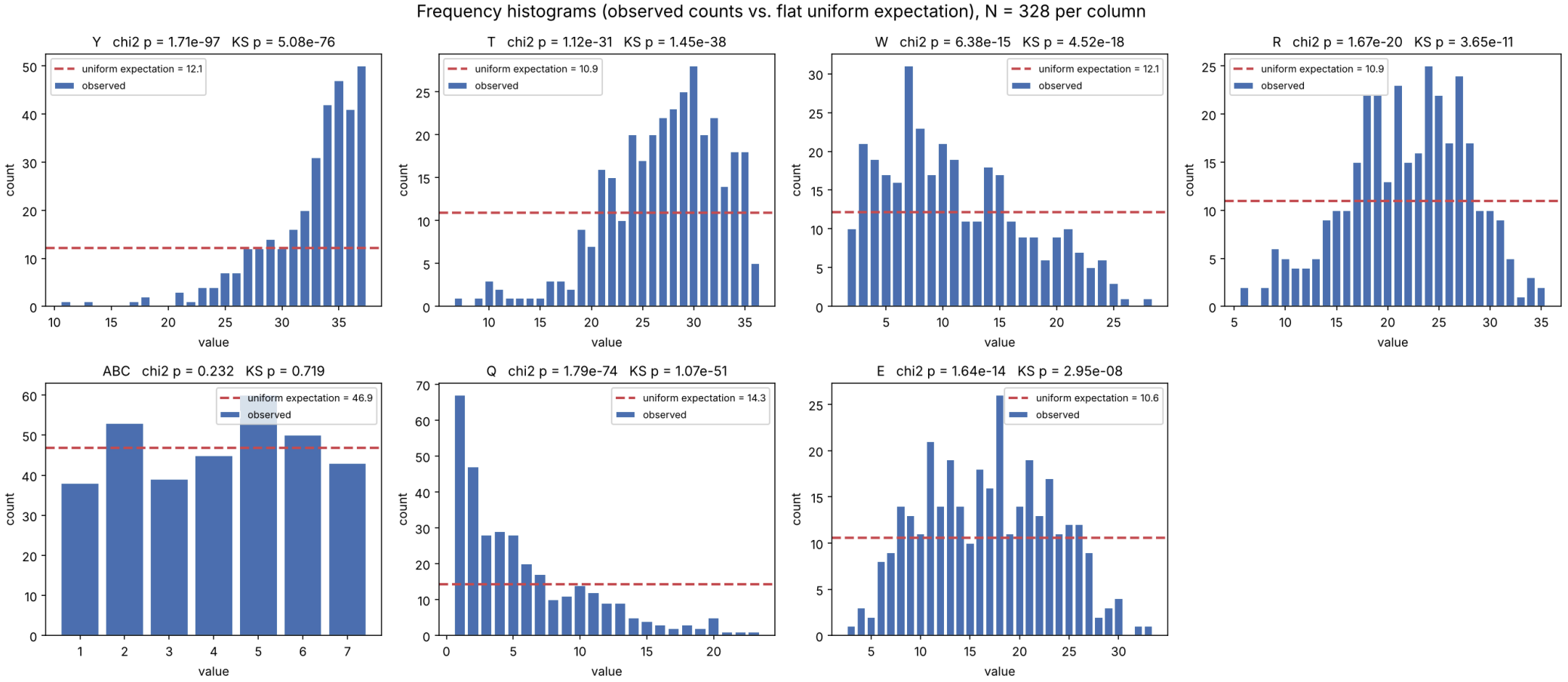
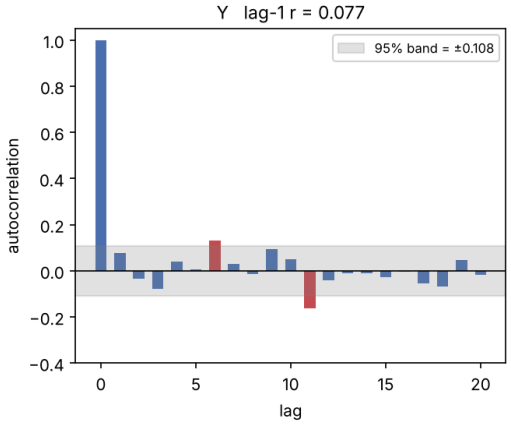
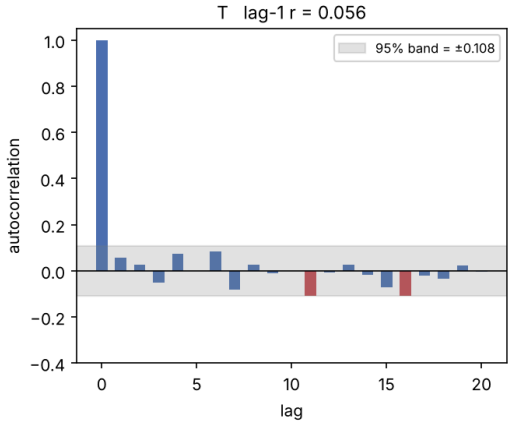
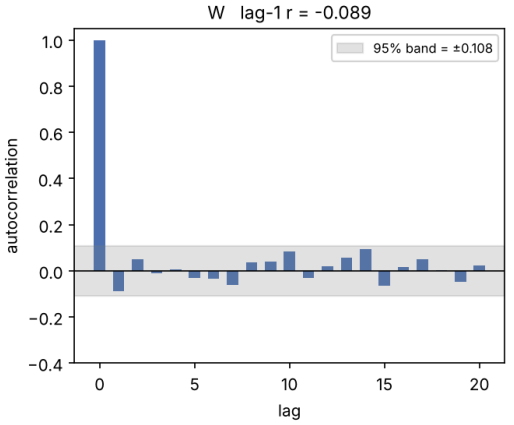


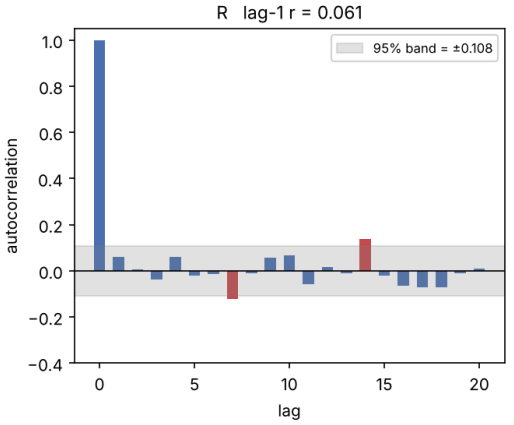
Figure 2: Correlograms (lags 0–20, 95% band)

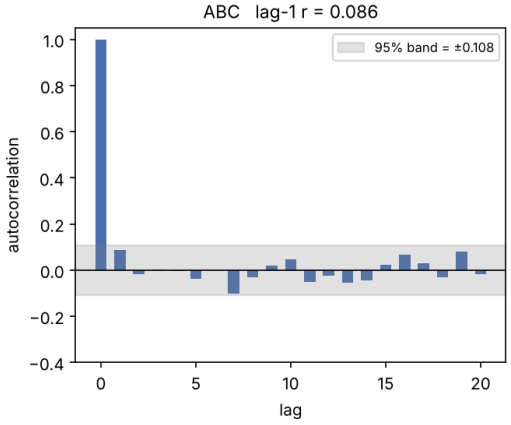
Correlograms, lags 0-20 (red = outside the 95% confidence band)

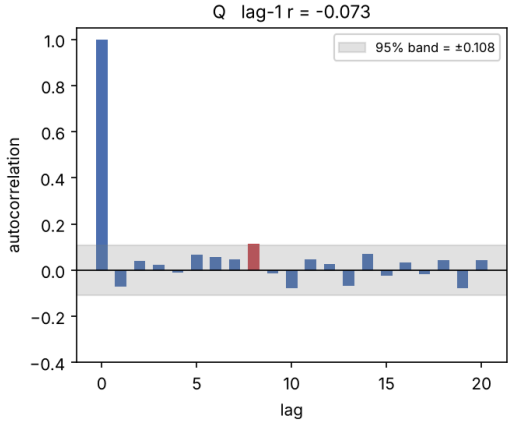


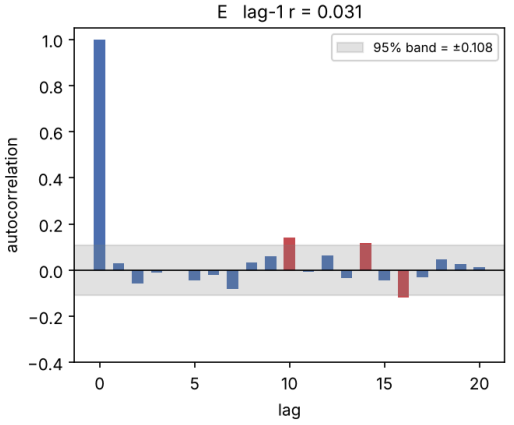












Appendix A: Method (fixed before results)

- Merge: the 7 files were aligned by row position (328 rows each, no missing values) → data/merged/merged_7_files.csv. SHA-256 hashes of the inputs are in README.md.
- Uniformity: Pearson χ^2 against the discrete uniform on [min, max] (expected count per value is 10.6–46.9, so every expected count is ≥ 5). Discrete KS (Conover 1972) with a conservative Kolmogorov p-value.
- Independence: sample ACF at lags 1–20 with a ±1.96/√N band. Runs up-and-down test (E=(2n−1)/3, Var=(16n−29)/90, zero differences dropped). Wald–Wolfowitz runs above/below the median (ties dropped). Both use the two-sided normal approximation.
- Entropy: plug-in Shannon entropy in bits, divided by $\log_2 k$. Binary entropy of each bit stream.
- NIST: Python implementation of SP 800-22 Rev.1a (scripts/nist_sp800_22.py), because the official NIST STS download host was not reachable from the analysis environment. Parameters: Block Frequency M = smallest value with M≥20, M>0.01n, N<100. Longest Run M=8 (n<6272) or 128. Non-overlapping template B=000000001, N=8 (all 148 aperiodic m=9 templates also run, see Appendix C). Serial m = $\lfloor \log_2 n \rfloor - 3$. ApEn m = $\lfloor \log_2 n \rfloor - 6$. α = 0.01.

Appendix B: Implementation validation against the NIST worked examples

```
VALIDATION AGAINST NIST SP 800-22 Rev.1a WORKED EXAMPLES
example                                          computed       NIST  match(6dp)
2.1.4 Frequency, e=1011010101                    0.527089   0.527089  YES
2.1.8 Frequency, 100 bits of pi                  0.109599   0.109599  YES
2.2.8 BlockFrequency M=10, 100 bits of pi        0.706438   0.706438  YES
2.3.4 Runs, e=1001101011                         0.147232   0.147232  YES
2.3.8 Runs, 100 bits of pi                       0.500798   0.500798  YES
2.4.8 LongestRun M=8, 128-bit example            0.180609   0.180609  YES
2.6.8 DFT, 100 bits of pi                        0.646355   0.168669  NO
2.7.4 NonOverlapping B=001 N=2, e=10100100101110010110   0.344154   0.344154  YES
2.11.4 Serial m=3, e=0011011101 (P1)             0.808792   0.808792  YES
2.11.4 Serial m=3, e=0011011101 (P2)             0.670320   0.670320  YES
2.12.8 ApEn m=2, 100 bits of pi                  0.235301   0.235301  YES
2.13.8 CuSum forward, 100 bits of pi             0.219194   0.219194  YES
2.13.8 CuSum backward, 100 bits of pi            0.114866   0.114866  YES
ALL SPEC EXAMPLES REPRODUCED: False
NOTE: a 'NO' row is reported as-is. For DFT the implementation follows the formulas of SP 800-22
sections 2.6.4/3.6 literally (moduli of the first n/2 DFT coefficients, T=sqrt(ln(1/0.05)n), N0=0.95n/2);
the counts N1 printed in the spec examples could not be reproduced from those formulas, so DFT results
in this study carry the flag 'implementation not validated against spec example'.
```

Appendix C: Non-overlapping template, all 148 templates (count with p ≥ 0.01)

stream	templates	passing	min_p
ABC	148	143	1.124e-05
COMBINED	148	136	3.719e-06
E	148	139	3.658e-05
Q	148	127	4.02e-59
R	148	142	0.0030
T	148	128	5.87e-10
W	148	136	1.254e-13
Y	148	133	2.109e-31

Appendix D: Raw machine output (results/machine_output.txt, unedited)

```
Merged table: 328 rows x 7 columns ['Y', 'T', 'W', 'R', 'ABC', 'Q', 'E']

=== 1. UNIFORMITY ===
column  min  max  k   N  expected_per_value  min_observed  max_observed  empty_values       chi2  df       p_chi2     KS_D         p_KS  chi2_p_gt_0_01  chi2_p_gt_0_05  KS_p_gt_0_01  KS_p_gt_0_05
     Y   11   37 27 328           12.148148             0            50             6 540.115854  26 1.706636e-97 0.498984 5.080926e-76           False           False         False         False
     T    7   36 30 328           10.933333             0            28             1 222.243902  29 1.124169e-31 0.359959 1.450363e-38           False           False         False         False
     W    2   28 27 328           12.148148             0            31             1 125.073171  26 6.376225e-15 0.246725 4.520667e-18           False           False         False         False
     R    6   35 30 328           10.933333             0            25             1 162.426829  29 1.673742e-20 0.192886 3.654623e-11           False           False         False         False
   ABC    1    7  7 328           46.857143            38            60             0   8.085366   6 2.319157e-01 0.037892 7.192103e-01            True            True          True          True
     Q    1   23 23 328           14.260870             1            67             0 416.274390  22 1.787199e-74 0.415164 1.066348e-51           False           False         False         False
     E    3   33 31 328           10.580645             0            26             1 130.573171  30 1.644915e-14 0.164831 2.953836e-08           False           False         False         False
Proportion of columns passing chi2 at alpha=0.01: 0.1429  (NIST acceptable range for m=7: [0.8772, 1.0000])
Proportion of columns passing KS   at alpha=0.01: 0.1429

=== 2. INDEPENDENCE ===
column  acf_lag1  max_abs_acf_lags1_20  band_95  lags_outside_band which_lags_outside  runs_updown_R  runs_updown_expected  runs_updown_z  runs_updown_p  runs_updown_zero_diffs_dropped  runs_median_R  runs_median_expected
runs_median_z  runs_median_p  median  n_above  n_below  ties_at_median_dropped  updown_p_gt_0_05  median_p_gt_0_05
     Y  0.076785              0.161015 0.108221                 2              6 11             190             189.666667       0.046979       0.962530                              43            144            143.825175
0.020737       0.983455    34.0      138      148                      42              True              True
     T  0.055578              0.110086 0.108221                 2             11 16             208             206.333333       0.225166       0.821850                              18            150            153.459016
-0.396886       0.691452    28.0      150      155                      23              True              True
     W -0.089353              0.094933 0.108221                 0                                 219             206.333333       1.711260       0.087033                              18            157            154.498371
0.286022       0.774861    10.0      153      154                      21              True              True
     R  0.061490              0.137473 0.108221                 2              7 14             218             211.000000       0.935136       0.349718                              11            140            157.370607
-1.968477       0.049013    22.0      161      152                      15              True             False
   ABC  0.085874              0.103369 0.108221                 0                                 191             188.333333       0.377166       0.706051                              45            134            141.565371
-0.907041       0.364385     4.0      153      130                      45              True              True
     Q -0.072586              0.114761 0.108221                 1               8             209             190.333333       2.626186       0.008635                              42            158            150.123746
0.914834       0.360279     4.0      157      142                      29             False              True
     E  0.031135              0.141132 0.108221                 3           10 14 16            205             210.333333      -0.713618       0.475464                              12            155            156.993590
-0.226103       0.821122    17.0      155      157                      16              True              True
Expected number of lags outside the 95% band among 20 under independence: 1.0 per column

=== 3. ENTROPY ===
  column    k   H_bits    H_max  H_ratio meets_99_9  bits_per_value  n_bits  ones_fraction  H_binary  H_binary_meets_99_9  ref_ones_fraction_if_exactly_uniform  ref_expected_H_ratio_if_uniform_N328
       Y 27.0 3.669035 4.754888 0.771634      False             5.0    1640       0.553049  0.991865                False                                 0.444444                              0.987974
       T 30.0 4.342126 4.906891 0.884904      False             5.0    1640       0.560976  0.989245                False                                 0.473333                              0.987002
       W 27.0 4.442115 4.754888 0.934221      False             5.0    1640       0.397561  0.969506                False                                 0.444444                              0.987974
       R 30.0 4.516651 4.906891 0.920471      False             5.0    1640       0.504878  0.999931                 True                                 0.473333                              0.987002
     ABC  7.0 2.789867 2.807355 0.993771      False             3.0     984       0.434959  0.987759                False                                 0.428571                              0.995300
       Q 23.0 3.775553 4.523562 0.834642      False             5.0    1640       0.272561  0.845116                False                                 0.417391                              0.989304
       E 31.0 4.608076 4.954196 0.930136      False             5.0    1640       0.485366  0.999382                 True                                 0.483871                              0.986683
COMBINED  NaN      NaN      NaN      NaN        NaN             NaN   10824       0.459904  0.995356                False                                      NaN                                   NaN

=== 4. NIST SP 800-22 (15 tests) ===
  stream  n_bits                   test  sub       p_value  result p_gt_0_05
       Y    1640              Frequency    1  1.734192e-05 FAILURE     False
       Y    1640          BlockFrequency    1  8.775571e-01 SUCCESS      True
       Y    1640                   Runs    1  0.000000e+00 FAILURE     False
       Y    1640              LongestRun    1  1.120545e-06 FAILURE     False
       Y    1640        BinaryMatrixRank    0           NaN     N/A      None
       Y    1640                    DFT    1  3.221110e-03 FAILURE     False
       Y    1640  NonOverlappingTemplate    1  9.305624e-01 SUCCESS      True
       Y    1640     OverlappingTemplate    0           NaN     N/A      None
       Y    1640         MaurerUniversal    0           NaN     N/A      None
       Y    1640        LinearComplexity    0           NaN     N/A      None
       Y    1640                  Serial    1  6.499631e-36 FAILURE     False
       Y    1640                  Serial    2  1.081033e-06 FAILURE     False
       Y    1640      ApproximateEntropy    1  1.632328e-24 FAILURE     False
       Y    1640          CumulativeSums    1  2.772722e-05 FAILURE     False
       Y    1640          CumulativeSums    2  3.102013e-05 FAILURE     False
       Y    1640        RandomExcursions    0           NaN     N/A      None
       Y    1640 RandomExcursionsVariant    0           NaN     N/A      None
       T    1640              Frequency    1  7.866607e-07 FAILURE     False
       T    1640          BlockFrequency    1  8.586509e-01 SUCCESS      True
       T    1640                   Runs    1  0.000000e+00 FAILURE     False
       T    1640              LongestRun    1  1.714225e-03 FAILURE     False
       T    1640        BinaryMatrixRank    0           NaN     N/A      None
       T    1640                    DFT    1  1.000000e+00 SUCCESS      True
       T    1640  NonOverlappingTemplate    1  9.305624e-01 SUCCESS      True
       T    1640     OverlappingTemplate    0           NaN     N/A      None
```

```
      T    1640         MaurerUniversal    0           NaN     N/A      None
      T    1640        LinearComplexity    0           NaN     N/A      None
      T    1640                  Serial    1  1.756582e-08 FAILURE     False
      T    1640                  Serial    2  4.085053e-02 SUCCESS     False
      T    1640      ApproximateEntropy    1  2.946622e-11 FAILURE     False
      T    1640          CumulativeSums    1  8.292651e-07 FAILURE     False
      T    1640          CumulativeSums    2  1.573321e-06 FAILURE     False
      T    1640        RandomExcursions    0           NaN     N/A      None
      T    1640 RandomExcursionsVariant    0           NaN     N/A      None
      W    1640               Frequency    1  1.068375e-16 FAILURE     False
      W    1640           BlockFrequency    1  2.260117e-02 SUCCESS     False
      W    1640                    Runs    1  0.000000e+00 FAILURE     False
      W    1640             LongestRun    1  1.275836e-14 FAILURE     False
      W    1640        BinaryMatrixRank    0           NaN     N/A      None
      W    1640                     DFT    1  6.763008e-04 FAILURE     False
      W    1640  NonOverlappingTemplate    1  7.095561e-01 SUCCESS      True
      W    1640     OverlappingTemplate    0           NaN     N/A      None
      W    1640         MaurerUniversal    0           NaN     N/A      None
      W    1640        LinearComplexity    0           NaN     N/A      None
      W    1640                  Serial    1  8.548811e-17 FAILURE     False
      W    1640                  Serial    2  5.478578e-03 FAILURE     False
      W    1640      ApproximateEntropy    1  2.262213e-18 FAILURE     False
      W    1640          CumulativeSums    1  1.519082e-17 FAILURE     False
      W    1640          CumulativeSums    2  5.341877e-17 FAILURE     False
      W    1640        RandomExcursions    0           NaN     N/A      None
      W    1640 RandomExcursionsVariant    0           NaN     N/A      None
      R    1640               Frequency    1  6.927751e-01 SUCCESS      True
      R    1640           BlockFrequency    1  9.998708e-01 SUCCESS      True
      R    1640                    Runs    1  1.209568e-09 FAILURE     False
      R    1640             LongestRun    1  3.468172e-04 FAILURE     False
      R    1640        BinaryMatrixRank    0           NaN     N/A      None
      R    1640                     DFT    1  1.000000e+00 SUCCESS      True
      R    1640  NonOverlappingTemplate    1  9.305624e-01 SUCCESS      True
      R    1640     OverlappingTemplate    0           NaN     N/A      None
      R    1640         MaurerUniversal    0           NaN     N/A      None
      R    1640        LinearComplexity    0           NaN     N/A      None
      R    1640                  Serial    1  1.979151e-06 FAILURE     False
      R    1640                  Serial    2  1.336046e-02 SUCCESS     False
      R    1640      ApproximateEntropy    1  9.212154e-07 FAILURE     False
      R    1640          CumulativeSums    1  9.975289e-01 SUCCESS      True
      R    1640          CumulativeSums    2  7.787957e-01 SUCCESS      True
      R    1640        RandomExcursions    0           NaN     N/A      None
      R    1640 RandomExcursionsVariant    0           NaN     N/A      None
    ABC     984               Frequency    1  4.494068e-05 FAILURE     False
    ABC     984           BlockFrequency    1  8.915846e-01 SUCCESS      True
    ABC     984                    Runs    1  0.000000e+00 FAILURE     False
    ABC     984             LongestRun    1  5.233210e-05 FAILURE     False
    ABC     984        BinaryMatrixRank    0           NaN     N/A      None
    ABC     984                     DFT    0           NaN     N/A      None
    ABC     984  NonOverlappingTemplate    1  5.935664e-01 SUCCESS      True
    ABC     984     OverlappingTemplate    0           NaN     N/A      None
    ABC     984         MaurerUniversal    0           NaN     N/A      None
    ABC     984        LinearComplexity    0           NaN     N/A      None
    ABC     984                  Serial    1  1.052980e-02 SUCCESS     False
    ABC     984                  Serial    2  5.228529e-01 SUCCESS      True
    ABC     984      ApproximateEntropy    1  7.243013e-09 FAILURE     False
    ABC     984          CumulativeSums    1  8.988135e-05 FAILURE     False
    ABC     984          CumulativeSums    2  6.818594e-05 FAILURE     False
    ABC     984        RandomExcursions    0           NaN     N/A      None
    ABC     984 RandomExcursionsVariant    0           NaN     N/A      None
      Q    1640               Frequency    1  8.888321e-76 FAILURE     False
      Q    1640           BlockFrequency    1  2.959868e-43 FAILURE     False
      Q    1640                    Runs    1  0.000000e+00 FAILURE     False
      Q    1640             LongestRun    1  2.417923e-43 FAILURE     False
      Q    1640        BinaryMatrixRank    0           NaN     N/A      None
      Q    1640                     DFT    1  1.470093e-08 FAILURE     False
      Q    1640  NonOverlappingTemplate    1  5.050061e-58 FAILURE     False
      Q    1640     OverlappingTemplate    0           NaN     N/A      None
      Q    1640         MaurerUniversal    0           NaN     N/A      None
      Q    1640        LinearComplexity    0           NaN     N/A      None
      Q    1640                  Serial    1 7.972201e-211 FAILURE     False
      Q    1640                  Serial    2  9.814274e-33 FAILURE     False
      Q    1640      ApproximateEntropy    1  1.949306e-68 FAILURE     False
      Q    1640          CumulativeSums    1  4.444161e-76 FAILURE     False
      Q    1640          CumulativeSums    2  4.444161e-76 FAILURE     False
      Q    1640        RandomExcursions    0           NaN     N/A      None
      Q    1640 RandomExcursionsVariant    0           NaN     N/A      None
      E    1640               Frequency    1  2.359085e-01 SUCCESS      True
```

```
       E    1640          BlockFrequency    1  9.887851e-01 SUCCESS      True
       E    1640                    Runs    1  7.999725e-09 FAILURE     False
       E    1640              LongestRun    1  2.539215e-02 SUCCESS     False
       E    1640        BinaryMatrixRank    0           NaN     N/A      None
       E    1640                     DFT    1  1.126928e-01 SUCCESS      True
       E    1640  NonOverlappingTemplate    1  9.305624e-01 SUCCESS      True
       E    1640     OverlappingTemplate    0           NaN     N/A      None
       E    1640         MaurerUniversal    0           NaN     N/A      None
       E    1640        LinearComplexity    0           NaN     N/A      None
       E    1640                  Serial    1  8.205748e-03 FAILURE     False
       E    1640                  Serial    2  3.174829e-01 SUCCESS      True
       E    1640      ApproximateEntropy    1  1.301875e-06 FAILURE     False
       E    1640          CumulativeSums    1  4.520147e-01 SUCCESS      True
       E    1640          CumulativeSums    2  4.154880e-01 SUCCESS      True
       E    1640        RandomExcursions    0           NaN     N/A      None
       E    1640 RandomExcursionsVariant    0           NaN     N/A      None
COMBINED   10824               Frequency    1  7.238692e-17 FAILURE     False
COMBINED   10824          BlockFrequency    1  8.416109e-04 FAILURE     False
COMBINED   10824                    Runs    1  0.000000e+00 FAILURE     False
COMBINED   10824              LongestRun    1  2.748663e-24 FAILURE     False
COMBINED   10824        BinaryMatrixRank    0           NaN     N/A      None
COMBINED   10824                     DFT    1  1.214722e-05 FAILURE     False
COMBINED   10824  NonOverlappingTemplate    1  3.718841e-06 FAILURE     False
COMBINED   10824     OverlappingTemplate    0           NaN     N/A      None
COMBINED   10824         MaurerUniversal    0           NaN     N/A      None
COMBINED   10824        LinearComplexity    0           NaN     N/A      None
COMBINED   10824                  Serial    1  1.690416e-21 FAILURE     False
COMBINED   10824                  Serial    2  3.488394e-02 SUCCESS     False
COMBINED   10824      ApproximateEntropy    1  1.083137e-30 FAILURE     False
COMBINED   10824          CumulativeSums    1  3.336541e-17 FAILURE     False
COMBINED   10824          CumulativeSums    2  2.612509e-17 FAILURE     False
COMBINED   10824        RandomExcursions    0           NaN     N/A      None
COMBINED   10824 RandomExcursionsVariant    0           NaN     N/A      None

Non-overlapping template, all 148 aperiodic m=9 templates:
          templates  passing         min_p
stream
ABC             148      143  1.123558e-05
COMBINED        148      136  3.718841e-06
E               148      139  3.657943e-05
Q               148      127  4.019792e-59
R               148      142  2.973804e-03
T               148      128  5.869751e-10
W               148      136  1.253583e-13
Y               148      133  2.109432e-31

SCORECARD
                  test     combined_p_values  combined_min_p   proportion  m_applicable  prop_lower_bound status                                                
```